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In [1]: import pandas as pd
import numpy as np
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In [4]: df = pd.read_csv("Data Cleaning 12.08.csv")
print(df.shape)
display(df.head(1))
```

(5986, 10)

	name	planet_status	mass	mass_error_min	mass_error_max	mass_sini	mass_sini_error_m
0	109 Psc b	Confirmed	5.743	0.289	1.011	6.383	0.07

```
In [5]: df.describe()
```

	mass	mass_error_min	mass_error_max	mass_sini	mass_sini_error_min	mass_sini_error_max
count	2079.000000	2817.000000	2817.0000	1202.000000	1039.000000	1039.000000
mean	1.492538	0.324366	inf	1.983529	0.214687	0.214687
std	2.602286	1.351929	NaN	3.199717	0.636474	0.636474
min	0.000190	0.000000	-17.8000	0.000470	0.000000	0.000000
25%	0.029655	0.006290	0.0057	0.045000	0.005940	0.005940
50%	0.370000	0.041000	0.0400	0.731500	0.049300	0.049300
75%	1.466500	0.170000	0.1700	2.400000	0.170000	0.170000
max	12.934000	53.500000	inf	50.000000	13.000000	13.000000

```
In [6]: df.info()
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```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5986 entries, 0 to 5985
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   name                   5986 non-null   object
1   planet_status          5986 non-null   object
2   mass                   2079 non-null   float64
3   mass_error_min         2817 non-null   float64
4   mass_error_max         2817 non-null   float64
5   mass_sini              1202 non-null   float64
6   mass_sini_error_min    1039 non-null   float64
7   mass_sini_error_max    1039 non-null   float64
8   radius                 4568 non-null   float64
9   radius_error_min       4093 non-null   float64
dtypes: float64(8), object(2)
memory usage: 467.8+ KB
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In [7]: print(df.isnull().sum())
```

```
name                0
planet_status       0
mass                3907
mass_error_min      3169
mass_error_max      3169
mass_sini           4784
mass_sini_error_min 4947
mass_sini_error_max 4947
radius             1418
radius_error_min    1893
dtype: int64
```

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In [8]: df = df.drop_duplicates()
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In [9]: df = df.dropna()
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In [10]: df.select_dtypes(include="number").columns
```

```
Out[10]: Index(['mass', 'mass_error_min', 'mass_error_max', 'mass_sini',
               'mass_sini_error_min', 'mass_sini_error_max', 'radius',
               'radius_error_min'],
              dtype='object')
```

```
In [11]: print("Rows:", df.shape[0])
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Rows: 112

```
In [12]: print("Columns:", df.shape[1])
```

Columns: 10

```
In [13]: df.to_excel("cleaned_dataset.xlsx", index=False)
```